Limits of piecewise-defined functions

Recall that one can define the absolute value function $|x|$ as the following piecewise defined function:

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Example. To compute limits of piecewise-defined functions at a point where the function definition changes, one can use one-sided limits. For example,

$$\begin{aligned} \lim_{x\to 0^+} |x| &= \lim_{x\to 0^+} x \quad (\text{since } |x| = x \text{ for } x > 0) \\ &= \lim_{x\to 0} x \quad (\text{since } \lim_{x\to 0} x \text{ exists, } \lim_{x\to 0^+} x = \lim_{x\to 0^-} x = \lim_{x\to 0} x) \\ &= 0. \end{aligned}$$

Similarly,

$$\begin{aligned} \lim_{x\to 0^-} |x| &= \lim_{x\to 0^-} (-x) \quad (\text{since } |x| = -x \text{ for } x < 0) \\ &= -\lim_{x\to 0^-} x \quad (\text{the Limit Laws hold for one-sided limits}) \\ &= -\lim_{x\to 0} x \quad (\text{since } \lim_{x\to 0} x \text{ exists, } \lim_{x\to 0^+} x = \lim_{x\to 0^-} x = \lim_{x\to 0} x) \\ &= (-1)\cdot 0 \\ &= 0. \end{aligned}$$

Since $\lim_{x\to 0^-} |x| = 0 = \lim_{x\to 0^+} |x|$, $\lim_{x\to 0} |x| = 0$.

1. Find $\lim_{x\to 0^+} |x|$, $\lim_{x\to 0^-} |x|$, and $\lim_{x\to 0} |x|$, if these exist.

2. Find $\lim_{x\to 2^+} |x-2|$, $\lim_{x\to 2^-} |x-2|$, and $\lim_{x\to 2} |x-2|$, if these exist.

3. Find $\lim_{x\to 0^+} \frac{|x|}{x}$, $\lim_{x\to 0^-} \frac{|x|}{x}$, and $\lim_{x\to 0} \frac{|x|}{x}$, if these exist.

4. Find $\lim_{x\to 2^+}\frac{|x-2|}{2-x}$, $\lim_{x\to 2^-}\frac{|x-2|}{2-x}$, and $\lim_{x\to 2}\frac{|x-2|}{2-x}$, if these exist.

5. Let Let $f(x)=\begin{cases}4x-3 & \text{if } x<2,\\ (x-3)^2 & \text{if } x\geq 2.\end{cases}$. Find $\lim_{x\to 2^+} f(x)$, $\lim_{x\to 2^-} f(x)$, and $\lim_{x\to 2} f(x)$, if these exist.

Limits at infinity: polynomials and rational function

One can prove that

- $\lim\limits_{x\to\pm\infty} c = c$
- $\lim\limits_{x\to\infty} x^n = \infty$ for $n\in\mathbb{N}$
- $\lim\limits_{x\to-\infty} x^n = \begin{cases}\infty, & \text{if } n\in\mathbb{N} \text{ is even},\\ -\infty, & \text{if } n\in\mathbb{N} \text{ is odd}\end{cases}$
- $\lim\limits_{x\to\pm\infty}\frac{1}{x^n} = 0$ for $n\in\mathbb{N}$

These allow us to find limits at infinity for polynomials and rational functions.

Example.

$$\begin{aligned}\lim_{x\to-\infty}(5x^2-2x+2) &= \lim_{x\to-\infty} 5x^2(1-\frac{2}{5x}+\frac{2}{5x^2})\\ &= \lim_{x\to-\infty} 5x^2\cdot\lim_{x\to-\infty}(1-\frac{2}{5x}+\frac{2}{5x^2})\\ &= 5\lim_{x\to-\infty} x^2\cdot(\lim_{x\to-\infty}1-\frac{2}{5}\lim_{x\to-\infty}\frac{1}{x}+\frac{2}{5}\lim_{x\to-\infty}\frac{1}{x^2})\\ &= 5\cdot\infty(1-\frac{2}{5}\cdot 0+\frac{2}{5}\cdot 0)\\ &= \infty\cdot 1\\ &= \boxed{\infty}.\end{aligned}$$

6. Find $\lim_{x\to\infty}(2x^3-3x^2+5x+1)$ and $\lim_{x\to-\infty}(2x^3-3x^2+5x+1)$

Example.

$$\begin{aligned}
\lim_{x\to\infty}\frac{3x^3-2x^2+x+1}{7x^3-8x+4} &= \lim_{x\to\infty}\frac{x^3}{x^3}\cdot\frac{3-\frac{2}{x}+\frac{1}{x^2}+\frac{1}{x^3}}{7-\frac{8}{x^2}+\frac{4}{x^3}}\\
&= \lim_{x\to\infty}\frac{3-\frac{2}{x}+\frac{1}{x^2}+\frac{1}{x^3}}{7-\frac{8}{x^2}+\frac{4}{x^3}} \qquad \text{(factoring and canceling does not change the limit)}\\
&= \frac{\lim_{x\to\infty}3-2\lim_{x\to\infty}\frac{1}{x}+\lim_{x\to\infty}\frac{1}{x^2}+\lim_{x\to\infty}\frac{1}{x^3}}{\lim_{x\to\infty}7-8\lim_{x\to\infty}\frac{1}{x^2}+4\lim_{x\to\infty}\frac{}{x^3}}\\
&= \frac{3-2\cdot0+0+0}{7-8\cdot0+4\cdot0}\\
&= \boxed{\frac{3}{7}}.
\end{aligned}$$

7. Find $\lim_{x\to-\infty}\dfrac{3x^2-9}{2x^3+3x-1}$

Infinite limits: rational functions

One can prove that

- $\lim\limits_{x\to a^+} \dfrac{1}{(x-a)^n} = \infty$ for $n \in \mathbb{N}$
- $\lim\limits_{x\to a^-} \dfrac{1}{(x-a)^n} = \begin{cases} \infty, & \text{if } n \in \mathbb{N} \text{ is even,} \\ -\infty, & \text{if } n \in \mathbb{N} \text{ is odd.} \end{cases}$

One can use these to calculate infinite limits of rational functions.

Example.

$$\begin{aligned} \lim_{x\to 1^+} \frac{x^2-3}{(x-1)^3} &= \lim_{x\to 1^+} \left[(x^2-3) \frac{1}{(x-1)^3} \right] \\ &= \lim_{x\to 1^+} (x^2-3) \cdot \lim_{x\to 1^+} \frac{1}{(x-1)^3} \\ &= \lim_{x\to 1} (x^2-3) \cdot \lim_{x\to 1^+} \frac{1}{(x-1)^3} \\ &= (-2) \cdot \infty \\ &= \boxed{-\infty}. \end{aligned}$$

8. Find $\lim\limits_{x\to 1^+} \dfrac{x^2-3}{(x-1)^3}$.

Example. Make sure to factor and cancel first when using this method. For example,

$$\begin{aligned}\lim_{x\to 1^+}\frac{x^2-1}{(x-1)^3} &= \lim_{x\to 1^+}\left[(x^2-1)\frac{1}{(x-1)^3}\right]\\ &= \lim_{x\to 1^+}(x^2-1)\cdot\lim_{x\to 1^+}\frac{1}{(x-1)^3}\\ &= \lim_{x\to 1}(x^2-1)\cdot\lim_{x\to 1^+}\frac{1}{(x-1)^3}\\ &= 0\cdot\infty, \text{ indeterminate!}\end{aligned}$$

Instead,

$$\begin{aligned}\lim_{x\to 1^+}\frac{x^2-1}{(x-1)^3} &= \lim_{x\to 1^+}\frac{(x+1)(x-1)}{(x-1)^3}\\ &= \lim_{x\to 1^+}\frac{x+1}{(x-1)^2}\\ &= \lim_{x\to 1^+}\left[(x+1)\frac{1}{(x-1)^2}\right]\\ &= \lim_{x\to 1^+}(x+1)\cdot\lim_{x\to 1^+}\frac{1}{(x-1)^2}\\ &= \lim_{x\to 1}(x+1)\cdot\lim_{x\to 1^+}\frac{1}{(x-1)^2}\\ &= 2\cdot\infty\\ &= \boxed{\infty}.\end{aligned}$$

9. Find $\lim_{x\to 3^-}\dfrac{x^2-9}{(x-3)^4}$.